

1 Introduction:

This lecture provided an introduction to the theme of **Lexical Semantics**, which is the study of how words are modeled. We discussed the challenges associated with existing approaches and uncovered interesting insights while exploring the meaning of “meaning”.

2 Why Semantics?

In an Information Retrieval system, documents that exhibit strong token-level similarity with the query are retrieved as a response to the user. The concept of **semantic search**, however, emphasizes retrieving documents that are semantically relevant to the query, even if the exact words from the query do not appear in the text. Thus, semantic search provides users with resources that accurately address the **intent** behind their queries.

The usage of a **vector-space model** to represent documents and queries could, in principle, help define a notion of document semantics. However, this model assumes that the tokens defining the space are strictly orthogonal. Further, the vector-based notation treats documents as linear combinations of unrelated tokens and does not actually take into account the “meaning” of the documents.

As a result, we need to move beyond a meaning-agnostic representation of documents and develop methods to genuinely model the semantics of our corpus.

3 The Meaning of “Meaning”

The first bottleneck in modeling word meanings is that the very definition of “meaning” is vague. In the context of **programming languages**, we encounter three primary notions of meaning:

- **Denotational semantics** - What does a word denote or allude to in terms of well-defined objects?
- **Axiomatic semantics** - This conveys meaning by providing a logical description of the system’s state before and after a word is used.
- **Operational semantics** - This describes a word’s meaning based on the specific operation it is designed to perform.

While programming languages are restrictive when compared to natural language, these notions provide ideas worth thinking about.

The next issue in the expression of meaning arises when we depend on **language** to express a word’s meaning, because then we have to deal with the perceptual grounding problem of language,

which is inherently not self-contained.

Another important point that was brought up is that the meaning of a word in isolation is of little practical value - the **context** in which a word is used is important for deciphering its underlying semantics. Historically, there have been three main approaches to dealing with word meanings:

- **Referential meaning:** A word's meaning is described using real-world objects and representations (e.g., explaining the word "apple" using a physical apple).
- **Mental model meaning:** A word's meaning is described using abstract concepts and ontologies when real-world referencing is difficult (e.g., explaining the concept of "justice").
- **Wittgenstein's Language as Use theory** - a word has little meaning or value of its own, and its usage in a given context determines its meaning. This suggests that deriving a word's meaning through bottom-up distributional semantics - analyzing its various uses across texts - is more valuable than assigning a static definition to it. However, this approach is challenging because a word can appear in highly varied contexts, and there may not always be sufficient data to fully grasp its usage in a specific context.

Wittgenstein's theory indicates that linguistic meaning is strongly tied to human memory and experience. A single word can be interpreted differently by individuals possessing different mental models of its usage. Therefore, the receiver's mental model must be taken into account when communicating meaning.

This relationship between **language and memory** was studied by the cognitive theorist Roger Schank, who believed that language understanding is a memory-based process, and that we interpret information not based on innate grammatical structure - function correlations, but through organized memories and experiences. This is in contrast to the thoughts of Noam Chomsky, who reduced the problem of language understanding to one of structure and grammar.

To contrast between these two models of information extraction, let us consider the example of an article in a newspaper regarding a plane crash.

According to Chomsky's approach, the way an agent might go about trying to understand the article is by breaking the text down into syntactic elements and processing the structure and function of each element, integrating all such information to digest the meaning of the article. However, Schank argued that a typical human would go about this process differently. As soon as a human reads the headline of the article, which talks of a plane crash, the human would create expectations and hypothesize ideas regarding the details of the crash, and develop targeted doubts to verify if this preliminary visualization of the event is accurate. This theory agrees with cognitive observations that human efforts of searching and curiosity are heavily tuned to satisfy our perceptual apparatus.

An interesting point raised in class was that the meanings of many words become much clearer when their **opposites** are defined. For instance, the concept of "good" becomes sharper only when we understand what is "bad". This idea stems from the reductionist approach, which believes that the undivided world is tough to comprehend and analyze, and introducing structured divisions and

analyzing smaller components makes the world easier to analyze. Language is ultimately more effective when it establishes boundaries to convey an experience or denote a change.

What exactly do we mean when we say we have **understood** a word? One way to evaluate this is by examining the word's structure-function mapping. For example, if a robot is taught the command "jump," is subsequently asked to jump, and successfully executes the action, we can conclude it has understood the word. This method, however, is difficult to implement in cases where deep understanding and complex structure-function relationships are involved. For example, how can we verify if the robot has understood the meaning of abstract concepts?

4 Conclusion

This lecture covered some approaches to lexical semantics, and discussed both the rationales and limitations of several such approaches. The following classes will formalize this intuition and build towards the ideal of language understanding.